MAE3405 Aerospace Propulsion

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Topic 3-4 Problem Set Questions

Question 4

In an air standard Otto cycle the maximum and minimum temperatures are 1400ºC and 15ºC. The heat supplied to the cycle is 800 kJ/kg.

(i) Calculate the compression ratio and cycle thermal efficiency. [Ans: 5.24, 48.5%]
(ii) Calculate also the ratio of maximum to minimum pressures in the cycle assuming SSL as minimum cycle pressure. [Ans: 30.4]
(iii) What is the maximum possible thermal efficiency that could be obtained from the thermodynamic states that make up the cycle? [Ans: 82.8%]

Question 5

Methanol (CH_3OH) is a possible alternative SI engine fuel. For a stoichiometric fuel/air mixture, determine:

(i) the air to fuel ratio,
(ii) the LHV (kJ/kg) for gaseous fuel, and
(iii) the percentage reduction in LHV assuming the fuel to be liquid.

[Ans: (i) 6.43; (ii) 21.12 MJ/kg; (iii) 5.54%]

Question 6A

A 6-cylinder SI aircraft engine operates at SSL on the fuel-air Otto cycle. Its total swept volume is 3.0 Litres, and the compression ratio is 9.33. Assume the fuel is gaseous isooctane and the A/F ratio is stoichiometric. Calculate:

(i) The LHV (kJ/kg) of the fuel
(ii) The fuel-air cycle thermal efficiency and
(iii) The cycle MEP.

[Ans: (i) 44.88 MJ/kg, (ii) 43.9%, (iii) 1.76 MPa]

Question 7

Measurement of the exhaust from a piston engine operating on a hydrocarbon fuel reveals the following gas concentrations (volume basis):
CO_2 9.14%, CO 5.1%, H_2O 13.7% and H_2 2.31%. Nitrogen makes up the balance.

The mass flow rate of the exhaust is 0.1 kg/sec. Determine:

(i) the mass flow rate (kg/s) of each of the exhaust species,
(ii) the fuel flow rate and air to fuel ratio and
(iii) a possible fuel molecule in the form C_nH_{2n+2} (find n)

[Ans: (ii) A/F=12.57; (iii) n=7.9]

Question 8

For the 4-cylinder Lycoming 0-360 engine operating on aviation gasoline (avgas table A.3 chapter 1) determine the following for this at SSL:

(i) The air standard thermal efficiency of the engine. [Ans: 57.2%]

(ii) The valve clearance space. [Ans: 0.197 litres/cylinder]

(iii) The BMEP for take-off and the 65% cruise condition. [Ans: 944 kPa; 758 kPa]

(iv) The brake thermal efficiency of the engine at take-off assuming an air to fuel ratio of 14.7 and volumetric efficiency of 85%. [Ans: 30.6%]

Question 9

For the 4 cylinder, 4-stroke aircraft engine below, determine its power output and brake thermal efficiency. Assume the fuel is liquid isooctane.

Quantity	**Value**	**Quantity**	**Value**
IMEP	1.0MPa	Volumetric efficiency	78%
Bore	0.1m	Mechanical Efficiency	85%
Stroke	0.12m	Altitude	3000m
RPM	2500	True airspeed	175kph
Air to fuel ratio	14.8		

[Ans: 66.8 kW, 40%]

Question 10

The engine dynamometer test results for a 4-stroke air-cooled 4-cylinder horizontally opposed spark ignition aircraft engine operating on avgas ($(A/F)_s$=14.7) are listed in the table below.

Speed RPM	**Brake Load N**	**Air Litres/min**	**Fuel L/min**
2000	525	1905	0.210
2200	575	2201	0.250
2400	540	2255	0.260
2600	522	2355	0.275
2700	506	2360	0.278

The dynamometer used has a torque arm radius of 0.381m. The test cell conditions on the day of the test were a temperature of 21.1C, and pressure of 101.3kPa. The cylinder bore is 82.6mm, and stroke 102mm. Calculate and plot against speed:

(i) torque

(ii) brake power

(iii) SFC

(iv) BMEP

(v) volumetric efficiency

(vi) brake thermal efficiency

(vii) air to fuel ratio, and

(viii) equivalence ratio.

[Ans: At 2000 RPM (i) Q=200 Nm, (ii) $\dot{W}$=41.9 kW, (iii) SFC = 0.213 kg/kW-hr, (iv) BMEP = 1150 kPa, (v) η_v = 87%, (vi) η_{bth} = 38.7%, (vii) A/F=15.38, (viii) ϕ = 0.956. Ask Demonstrators to check answers at higher RPM.]

Question 11

For the engine in Question 10;

(i) Assuming the linear relationship between power and density in the notes (topic 1.2), construct a plot of brake power at 2700RPM against altitude up to 8,000m.

(ii) Determine the maximum ceiling of an aircraft using this engine for an equivalent airspeed of 150knots and ceiling altitude drag force 350N. You may also assume a constant propulsive efficiency of 85%.

[Ans: (i) eg z=2000m, 43.5kW; (ii) ~3250m]